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The impact of returns policies on profitability: A fashion e-commerce case☆

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ABSTRACT

Research on returns policies is scarce and inconclusive. This study is a first attempt to empirically analyze and describe the effects of returns policies on consumer behavior and the moderating effects of the policies on profitability. The method involves analyzing the transactional data of a Swedish online fashion retailer. The results of the regression analyses show that with a lenient returns policy, repeat customers generate a significantly higher contribution per order, while returners and customers who enjoy free returns generate a significantly lower contribution per order. In addition, returners and repeat customers generate a significantly higher total contribution, while customers who enjoy free returns generate a significantly lower total contribution. Hence, returns policies that are free of charge do not necessarily benefit retailers in terms of long-term profitability. From a managerial perspective, the results provide guidance on how to address the recent changes in European consumer legislation.

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1. Introduction

A trend towards more liberalized or lenient returns policies is evident in the fashion e-commerce business. We may consider that retailers use the returns policy as a tool to reduce consumer risk and increase consumer demand (Janakiraman, Syrdal, & Freling, 2015). Retailers use returns policy in different ways: pre-purchase, to signal a certain level of retailer or product quality (Bonifield, Cole, & Schultz, 2010; Mukhopadhyay & Setaputra, 2007), and post-purchase, to alter product evaluations (Kim & Wansink, 2012). Returns policies can have varying degrees of leniency; Janakiraman et al. (2015) classify five returns policy leniency dimensions: time, money, effort, scope, and exchange.

Many markets do not have a law that requires retailers to accept returns (Chen & Grewal, 2013). In the European Union (EU), the law entitled consumers (Directive 97/7/EC, 1997) to return what they purchase online. Certain countries, such as Finland and Germany, interpreted the previous Distance Selling Directive (Directive 97/7/EC, 1997) in a way that allows customers to return products without having to pay for the return postage (monetary leniency). However, this directive has recently changed: by Jun. 13, 2014, all member states have implemented a new directive (Directive 2011/83/EU, 2011), which no longer requires organizations in the EU to allow their customers to return products free of charge (European Parliament, 2011). With this recent change, e-commerce organizations in Sweden and other EU nations that previously accepted free returns when selling online in Finland and Germany can now decide whether to continue to do so or to shift the cost of returns to consumers or segments of consumers, as Bonifield et al. (2010) suggest.

Opinions differ regarding returns policies and their level of restriction (Bahn & Boyd, 2014). In addition, the research is inconclusive on the effects of these policies (Janakiraman et al., 2015). Research indicates that returns policies used for short-term gain can have long-term negative consequences for retailers (Bower & Maxham, 2012). Retailers may use lenient return polices based on the belief that these policies increase purchases more than returns, despite the cost of processing returns (Janakiraman et al., 2015). This study sheds light on these issues and evaluates both the short- (order) and long-term (repurchase) effects of returns policy leniency on one dimension at a time, as Janakiraman et al. (2015) suggest, albeit the study is not a controlled field study.

The empirical study presented in this paper examines the impact of returns policies on consumer purchase and return behaviors (habits) in e-commerce, and the moderating effects of returns policies on profitability. The study examines the effects in a European context and addresses the recent changes in the European directive (Directive 2011/83/EU, 2011) on consumer rights that was implemented in a national legislation in the EU in June 2014. Since the legislation is relatively new, the results presented in this paper can assist managers in deciding whether to continue to offer lenient policies or to segment customers and offer differentiated policies, as researchers (Bonifield et al., 2010) suggest.

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This research continues a trend in the recent literature that use and analyze transactional data from the e-commerce industry (Bower & Maxham, 2012; Griffis, Rao, Goldsby, & Niranjan, 2012; Hjort, Lantz, Ericsson, & Gattorna, 2013; Lantz & Hjort, 2013). The aim of the study is to complement the existing literature by adding the complexity of returns policies to the analysis of the profitability of consumer purchase and return habits. This study also examines whether the conclusions of Lantz and Hjort (2013) are valid outside of their randomized and controlled experiment: that retailers should probably not offer free returns policies to all or some of the customers given that such policies are not mandatory from a legislative and/or competitive point of view. From a managerial perspective, this research intends to help guide organizations on how to set their returns policies with regard to the recent legislative changes.

The organization of the remainder of this paper is as follows. Section 2 presents the literature review, and Section 3 describes the methodology. Section 4 presents the results of the statistical analysis, and Section 5 discusses the results. Section 6 presents the conclusions and implications of the study.

2. Literature and theoretical background

2.1. Returns policy, buyer behavior, and profitability

We may consider the consumer's relationship with a retailer in an online setting as a series of transactions where the consumer evaluates the risks in terms of all the perceived transactions, and this series of transactions all boils down to a decision to purchase or not (Griffis et al., 2012). This concept implies that transaction cost economy theory can explain consumer-retailer transactions (Williamson, 1979). Chircu and Mahajan (2006) define a retail transaction as "an exchange between a consumer and a retailer in which the two parties obtain something from each other at a cost to each" (p. 899). Reducing risks or lowering transaction costs, such as by providing post-sale services and returns, can increase customer value (Chircu & Mahajan, 2006). Griffis et al. (2012) show that the customer's confidence in a retailer grows through a series of successful transactions, which includes returns processing.

The goal of lenient return conditions is often to create a marketing incentive (Rogers & Tibben-Lembke, 1999, Chapter 1) to attract and create loyal and repeat customers, and thus, to increase sales. However, a fundamental correlation between increasing sales and maximizing profitability does not exist, and profit is always the firm's first consideration (Yan, 2009). Lantz and Hjort (2013) show in their experiment that free returns are actually associated with a decrease in the average value of orders, an increased probability of returns, as well as an increased order frequency. Nevertheless, even though a liberal returns policy increases returns (Wood, 2001), a returns policy is still always valuable for e-commerce according to Yan (2009). Research on returns policies is still in development (Bower & Maxham, 2012; Janakiraman et al., 2015). An increasing stream of this research addresses returns policies in a B2C setting, and more specifically, in e-commerce with regard to consumer returns. Researchers often discuss the effects of these policies in terms of either profitability (Lantz & Hjort, 2013; Mukhopadhyay & Setaputra, 2007) or loyalty and behavioral aspects (Bonifield et al., 2010; Griffis et al., 2012; Hjort & Lantz, 2012; Pei, Paswan, & Yan, 2014; Wood, 2001). This research trend leaves a gap in the literature regarding how returns policies affect purchase and return behaviors and whether and/or how these policies affect profitability. The behavioral aspects often discuss pre- and post-returns spending in relation to different returns policies (Bower & Maxham, 2012). Knowledge regarding spending, behavior, loyalty, and purchase intentions is interesting and insightful, but one can argue that without a direct and empirical connection with profitability, these aspects are of more interest to academics than to practitioners.

Returns policies are clearly aspects of the online consumer experience, and liberal policies increase sales (i.e., demand) (Bower & Maxham, 2012; Mukhopadhyay & Setaputra, 2007; Petersen & Kumar, 2010). Griffis et al. (2012) suggest that the increased value of consumer loyalty can offset the incurred cost of a lenient returns policy. Another empirical study (Hjort et al., 2013), where repeat customers and non-returners, on average, generate a higher contribution per order, provides partial support for this finding. However, the total contribution of repeat customers per year is higher if they are also returners. Hjort et al. (2013), though, do not control for the degree of returns policy leniency.

Meanwhile, returns policies also decrease profitability. In a randomized and controlled experiment with real e-customers as participants, Hjort and Lantz (2012) confirm increased sales but conclude that, generally, it is not optimal in term of profitability to offer the same delivery and return conditions to all types of customers. Lantz and Hjort (2013) go on to argue that the main managerial implication of their study is to not offer free delivery and return policies, given that they are not required from a legislative and/or competitive point of view. They argue that an economic perspective would not recommend such policies due to the absence of a significant upside to compensate for the downside (i.e., a decreased coverage of costs).

Janakiraman et al. (2015) challenge what they describe as an "underlying assumption in most prior returns policy research — that all forms of returns policies affect product purchase and returns in a similar manner" (p. 8). They also suggest that returns policies per se benefit retailers, at least in increased product purchase. However, literature appears to report an inconclusive understanding of the effects of returns policies in prior research as Janakiraman et al. (2015) previously mention, or at least that prior research describes the effects only for one aspect at a time, that is, for either behavior/intention or profitability. This tendency might also explain why (as the introduction states) retailers use lenient returns policies based on the belief that such policies increase purchases more than returns, despite the cost of processing returns (Janakiraman et al., 2015). Consequently, this study analyzes the moderating effects of returns policies on consumer behavior and whether and/or how these affect profitability per customer by measuring the contribution per order and the total contribution over time, aggregating actual repurchase behavior, not intention (see Fig. 1). The study aims to determine whether and/or how the moderating effects vary between customers with and without experience with the retailer, as the literature states (Hernández, Jiménez, & Martín, 2010). The existing research indicates that customer behavior does not remain stable over time because past purchases influence subsequent behavior (Hernández et al., 2010).

The study's three hypotheses based on the theoretical and empirical literature reviewed above are as follows:

H1. Repeat customers are, on average, more profitable than non-repeat customers are.

H2. Non-returners are, on average, more profitable than returners are.

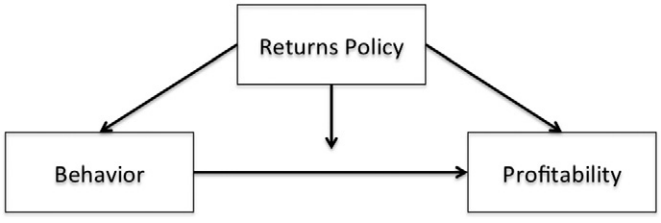


Fig. 1. Returns policy effects on consumer behavior and its moderating effects on profitability.

H3. Customers who do not enjoy free returns are, on average, more profitable than customers who do.

3. Methodology

The subject of this study is a Swedish online fashion retailer operating in Northern Europe. The company, which started operations in 2003, is one of the leading online fashion retailers in Northern Europe catering to female and male customers between the ages of 18 and 35. At the time of the research, the company offers two different returns policies (see Table A.1 in the Appendix A) depending on the location of order placement. For Danish, Norwegian, and Swedish consumers, the return freight costs approximately €4. In contrast, customers from other countries do not pay for the return freight. The company provided transactional data exported from its enterprise resource planning system for this study's quantitative analysis. The data contain those for all of the company's orders (1185,000 orders) for a period of two years across their markets in Denmark, Finland, Germany, The Netherlands, Norway, Sweden, and the UK.

The study conducts the analysis at the customer level and performs detailed calculations to reveal various aspects of each customer's behaviors, including the customer's order sales figures, return figures, and contribution per order. After that, the study analyzes each customer in terms of total sales, average sales per order, average contribution, total contribution, total number of orders, and total number of returns.

The organization's operations manager supplied us with vital information regarding the exported data set, freight costs, return freight costs, and costs related to order and return handling.

The study describes differences in contribution per order and total contribution per customer and further analyzes them using multiple regression analysis based on three explanatory variables:

- PH (purchase habits) — a binary variable indicating whether the customer is a repeat customer during the study period;
- RH (return habits) — a binary variable indicating whether the customer returns at least one item during the period; and
- FR (free returns) — a binary variable indicating whether the customer enjoys free returns during the period.

The study uses SPSS version 22 to perform all statistical analyses.

4. Results

4.1. Contribution per order

Table 1 presents the descriptive statistics regarding the average contribution per order for all combinations of PH, RH, and FR.

The study initially analyzes the observed differences in average contribution per order using the following regression model:

$$\text{AVCONT} = \beta_0 + \text{PH}\beta_1 + \text{RH}\beta_2 + \text{FR}\beta_3 + \varepsilon$$

where AVCONT is the average contribution (in SEK) per order. Table 2 shows the results of this regression. As expected, all factors are significant owing to the large sample size, and this result means that the study estimated the coefficients with very high accuracy, as the very small standard errors indicate. Repeat customers generate a significantly higher contribution per order ($B = 33.1$, $p < 0.001$). Returners and customers who enjoy free returns generate a significantly lower contribution per order ($B = -185.1$, $p < 0.001$ and $B = -66.4$, $p < 0.001$, respectively). Finally, the variance inflation factor (VIF) analysis shows no multicollinearity problems with the conducted regression, as all variance inflation factors are very small.

However, previous research (e.g. Hjort et al., 2013, Lantz & Hjort, 2013) indicate the presence of interaction effects between the factors.

Table 1
Average contribution (SEK) per order.

Purchase habits	Return habits	Free returns	Mean	SD	No. of orders (thousands)
		0	429	443	333
	0	1	387	398	139
		Total	417	431	472
		0	169	327	75
0	1	1	126	246	104
		Total	144	284	180
		0	381	436	408
	Total	1	275	365	244
		Total	342	414	652
		0	409	334	177
	0	1	342	322	39
		Total	397	333	217
		0	304	259	212
1	1	1	207	221	105
		Total	272	251	317
		0	352	300	389
	Total	1	244	260	144
		Total	323	294	533
		0	423	409	511
	0	1	378	383	179
		Total	411	403	689
		0	269	285	287
Total	1	1	167	237	209
		Total	226	270	496
		0	367	376	798
	Total	1	264	330	387
		Total	333	365	1185

In order to capture such interaction, the study analyzes the observed differences in average contribution per order using the following extended regression model:

$$\text{AVCONT} = \beta_0 + \text{PH}\beta_1 + \text{RH}\beta_2 + \text{FR}\beta_3 + (\text{PH} \times \text{RH})\beta_4 + (\text{PH} \times \text{FR})\beta_5 + (\text{RH} \times \text{FR})\beta_6 + \varepsilon.$$

Table 3 shows the results of this regression. While returners and customers who enjoy free returns still generate a significantly lower contribution per order ($B = -254.3$, $p < 0.001$ and $B = -38.4$, $p < 0.001$, respectively), repeat customers now also generate a significantly lower contribution per order ($B = -16.8$, $p < 0.001$). However, the results also indicate significant interaction effects between the examined factors. For repeat customers, the average contribution per order is significantly higher if the customers are also returners ($B = 145.4$, $p < 0.001$). Furthermore, customers who enjoy free returns generate a significantly lower contribution per order if they are also repeat customers ($B = -40.0$, $p < 0.001$) or returners ($B = -13.5$, $p < 0.001$). Finally, the VIF analysis shows no apparent multicollinearity problems with the conducted regression, as all variance inflation factors are relatively small.

4.2. Total contribution per customer

Table 4 presents the descriptive statistics regarding the total contribution per customer for all combinations of PH, RH, and FR. Note that the values for non-repeat customers are the same as in Table 1.

Table 2
Average contribution (SEK) per order explained by purchase habits (PH), return habits (RH), and return conditions (free returns; FR).

	B	S.E.	t	p	VIF
(Constant)	417.6	0.525	796.1	<0.001	
PH	33.3	0.696	47.8	<0.001	1.151
RH	−185.1	0.708	−261.4	<0.001	1.171
FR	−66.4	0.710	−93.5	<0.001	1.063

$F = 30{,}990$ ($p < 0.001$), $R = 0.270$.

Table 3
Average contribution (SEK) per order explained by purchase habits (PH), return habits (RH), return conditions (free returns; FR), and interactions.

	B	S.E.	t	p	VIF
(Constant)	428.3	0.594	720.7	<0.001	
PH	−16.8	0.970	−17.3	<0.001	2.255
RH	−254.3	1.256	−202.4	<0.001	3.720
FR	−38.4	1.041	−36.8	<0.001	2.310
PH * RH	145.4	1.467	99.1	<0.001	4.081
PH * FR	−40.0	1.547	−25.9	<0.001	2.472
RH * FR	−13.5	1.530	−8.8	<0.001	3.292

F = 17,530 (p < 0.001), R = 0.286.

The study initially analyzes the observed differences in total contribution per customer using the following regression model:

$$\text{TOTCONT} = \beta_0 + \text{PH}\beta_1 + \text{RH}\beta_2 + \text{FR}\beta_3 + \varepsilon$$

where TOTCONT is the total contribution (in SEK) per customer. Table 5 shows the results of this regression. As before, all factors are significant owing to the large sample size. Not surprisingly, repeat customers generate a significantly higher total contribution (B = 951.5, p < 0.001). Furthermore, returners also generate a significantly higher total contribution (B = 22.0, p < 0.001), while customers who enjoy free returns generate a significantly lower total contribution (B = −314.0, p < 0.001). Again, the VIF analysis shows no multicollinearity problems with the conducted regression, as all variance inflation factors are very small.

In order to capture possible interaction effects, the study analyzes the observed differences in total contribution per customer using the following extended regression model:

$$\text{TOTCONT} = \beta_0 + \text{PH}\beta_1 + \text{RH}\beta_2 + \text{FR}\beta_3 + (\text{PH}\times\text{RH})\beta_4 + (\text{PH}\times\text{FR})\beta_5 + (\text{RH}\times\text{FR})\beta_6 + \varepsilon.$$

Table 6 shows the results of this regression. Repeat customers still generate a significantly higher total contribution (B = 907.0, p < 0.001). However, returners now generate a significantly lower total contribution (B = −166.0, p < 0.001), and customers who enjoy free returns generate a significantly higher total contribution (B = 16.9, p < 0.001). Furthermore, the results indicate certain significant interaction effects between the examined factors. For repeat customers, the total contribution is significantly higher if they are also returners (B = 491.0, p < 0.001). Customers who enjoy free returns generate a significantly lower total contribution if they are also repeat customers (B = −540.6, p < 0.001) or returners (B = −192.9, p < 0.001). As before, the VIF analysis shows no apparent multicollinearity problems with the conducted regression, as all variance inflation factors are relatively small.

Table 4
Total contribution (SEK) per customer.

Purchase habits	Return habits	Free returns	Mean	SD	No. of orders (thousands)
		0	429	443	333
	0	1	387	398	139
		Total	417	431	472
		0	169	327	75
0	1	1	126	246	104
		Total	144	284	180
		0	381	436	408
	Total	1	275	365	244
		Total	342	414	652
		0	1286	1410	177
	0	1	943	1110	39
		Total	1224	1367	217
		0	1671	2305	212
1	1	1	872	1367	105
		Total	1407	2078	317
		0	1496	1958	389
	Total	1	892	1302	144
		Total	1333	1825	533
		0	727	993	511
	0	1	509	668	179
		Total	671	925	689
		0	1277	2094	287
Total	1	1	500	1051	209
		Total	950	1775	496
		0	925	1510	798
	Total	1	504	895	387
		Total	788	1355	1185

Table 5
Total contribution (SEK) per customer explained by purchase habits (PH), return habits (RH), and return conditions (free returns; FR).

	B	S.E.	t	p	VIF
(Constant)	453.1	1.871	242.2	<0.001	
PH	951.5	2.483	383.3	<0.001	1.151
RH	22.0	2.525	8.7	<0.001	1.171
FR	−314.0	2.531	−124.1	<0.001	1.063

F = 66,380 (p < 0.001), R = 0.379.

5. Discussion

Retailers often use lenient return policies based on the belief that such policies will increase purchases more than returns (Janakiraman et al., 2015). Bonifield et al. (2010) note that "given the interest in online purchase behavior, it is somewhat surprising that researchers have not extensively studied product returns" (p. 1058). Indeed, research on this area is scarce. This research is the first attempt to empirically analyze and describe the effects of returns policies on consumer behavior and the policies' moderating effects on profitability, using transactional data from a Swedish Fashion online retailer. From a managerial perspective, the results can help provide guidance on how to address the changes in consumer legislation (Directive 2011/83/EU, 2011) and the legal right to reconsider the previous law's mandatory lenient and free returns in certain markets (Finland and Germany). Consistent with findings from prior research (Janakiraman et al., 2015; Lantz & Hjort, 2013), the results from this research suggest that retailers should be careful in their choice of leniency factors, the levels of leniency, and the usage of such leniency factors. Using free returns (monetary leniency) for short-term benefits, such as to increase sales/demand, do not necessarily increase benefits for retailers in terms of profitability, which includes the associated costs of handling returns. Prior research suggests that a lenient returns policy decreases average order value and increases both order frequency and return probability. Consistent with these suggestions, the study finds initial support for all three hypotheses (see Table 1), as all three can affect contribution. However, repeat customers generate a lower average contribution per order but a higher total contribution. This finding can be the result of a decreased coverage of cost together with an increased return probability (order level) and increased order frequency (total contribution). One potential explanation for the decrease in average profitability at the order level

Table 6
Total contribution (SEK) per customer explained by purchase habits (PH), return habits (RH), return conditions (free returns; FR), and interactions.

	B	S.E.	t	p	VIF
(Constant)	412.0	2.109	195.3	<0.001	
PH	907.0	3.442	263.5	<0.001	2.255
RH	−166.0	4.458	−37.2	<0.001	3.720
FR	16.9	3.695	4.6	<0.001	2.310
PH * RH	491.0	5.207	94.3	<0.001	4.081
PH * FR	−540.6	5.491	−98.4	<0.001	2.472
RH * FR	−192.9	5.430	−35.5	<0.001	3.292

F = 37,661 (p < 0.001), R = 0.401.

from using free returns is related to Janakiraman et al.'s (2015) finding that consumers with minor product grievances are likely to return products when no cost is (for the returner) associated. This tendency diminishes the credited value of low-value products in particular.

Studying the profitability of online sales with regard to returns policies is a complex issue with inherent interaction effects, as Lantz and Hjort (2013) describe. Hence, analyzing the moderating effects of returns policies on consumer purchase and return behaviors produces completely different profitability results when studying interactions. This outcome is especially the case for repurchase behavior, which adds up over time to a total contribution per customer. For the repeat customers in this study, the average contribution per order and the total contribution are higher if they are also returners, and for repeat customers, the average contribution per order and the total contribution are lower if they also enjoy free returns. From these results, the only (financial) reason for allowing consumers to return products free of charge is to attract new customers (see Table 6), because customers who enjoy free returns are less profitable if they are either returners and repeat customers or both. This finding is in line with Lantz and Hjort's (2013) finding that free returns do not offer a significant upside to cover the increasing cost of returns. This study's finding also supports Bower and Maxham's (2012) finding that policy changes for short-term gain can have long-term negative consequences for retailers.

This research contributes to the scarce body of literature on returns policy research, especially empirical studies on the effects of lenient policies on profitability. Bonifield et al. (2010) found that retailers use returns policies in different ways to signal a certain level of quality to the consumers. High-end consumers may expect high-quality products when they see a lenient returns policy (Bonifield et al., 2010). For retailers, selling a typical product using different returns policies in different markets may send the wrong signals, and that approach could increase product returns. Janakiraman et al. (2015) advise a segmented solution rather than a blanket returns policy that applies to all customers. Bonifield et al. (2010) also suggest different policies according to customer segments. This study also supports a customized returns policy, as the evidence suggests that the use of a free returns policy can attract new customers (short-term gain) to the channel without negatively affecting profitability, as seen in Table 6, at least for products whose return is unlikely. The results clearly show that returns policies can benefit retailers (in the long term) since returners are the most profitable customers if they also are repeat customers and do not enjoy free returns. This finding is new and highly relevant for the company under study as well as other organizations considering the recent legislative changes regarding returns policies within the EU.

6. Conclusion

This study examines a previously unexplored area, namely the moderating effects of returns policies on profitability in fashion e-commerce. Its findings add new knowledge to the existing body of literature and provide support for earlier findings in both empirical and theoretical research. However, the connection between e-commerce profitability and the three explanatory variables this study addresses is complex and needs further exploration in empirical and theoretical research.

This study's findings have several implications. First, the evidence suggests that managers should be careful when implementing a lenient returns policy when doing so is not mandatory. Second, managers should reconsider the use of a blanket returns policy in markets that previously require a free returns policy, in regards to the recently altered consumer legislation (Directive 2011/83/EU, 2011). Third, managers should instead customize their returns policy according customer segments.

7. Limitations and future research

This study has some limitations. Empirically analyzing the moderating effects of returns policies on profitability is, of course, context-specific. Hence, the results for the company studied may be the product of its specific target customers and thus, are not generalizable. This warrants further research to determine whether these findings are valid in other contexts as well.

Appendix A

A.1. Returns policy of the retail company under study

Returns policy	Customer location	
	Denmark, Norway, and Sweden	All other countries
Does the merchant:		
Include a pre-printed shipping label?	Yes	Yes
Impose a time limit on returns?	14 days	14 days
Pay for all return shipping costs?	No	Yes
Charge restocking fees?	No	No
Refund original shipping and handling fees?	Yes	Yes
Provide the customer service contact information?	Yes	Yes
Require pre-authorization?	No	No

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